PEST MANAGEMENT DECISION GUIDE: GREEN AND YELLOW LIST

Tomato leafminer in tomato - Zambia

Phthorimaea absoluta (syn. Tuta absoluta) Belinda in Nyanja, Bemba, Tonga languages. Kamondo in Mambwe language



Damaged tomato caused by tomato leafminer. (Source: Metin GULESCI, Leaf Tobacco, Bugwood.org, CC BY-NC 3.0)



Tomato leaf with tunnels and larvae. (Source: Metin GULESCI, Leaf Tobacco, Bugwood.org, CC BY-NC 3.0)



Adult moth on tomato leaves. Small brown-silverish wings (1cm). Long antennae. (Source: Marja van der Straten, NVWA Plant Protection Service, Bugwood.org)

Prevention

- Plant seedlings free from infestation.
- Rogue, or destroy by burning, all infected crop residues and wild host plants, such as the Apple-of-Peru weed, to prevent carryover infestation of pests to new tomato plants.
- Avoid planting tomato if nearby tomato, potato, eggplants, and pepper plants got infested with *Tuta*, because the adult moth can fly and those crops are of the same family.
- Clean all equipment used in the transportation of tomatoes, such as boxes, crates and trucks, using soapy water to kill larvae.
- Avoid disposing of infested fruits at the markets, trading sites and farms. To prevent the spread of the pest, burn infested fruits and plant remains.

Monitoring

- Check for green to light pink larvae with a black head found inside the tunnels or right on top of the leaf.
- Larvae are 0.5-0.9mm long.
- White to cream necrotic mines on the upper and lower sides of the leaf lamina.
- Silky webs produced by the larvae that fold the leaf partially/
- Blighted and necrotic leaves on the plants are due to infestation.
- Puncture marks on the fruits and stem with dried black frass at larvae exit holes.
- Trap the adults with pheromone water traps (e.g. pherodis at 4 packets per acre supplied by Livestock services in Lusaka showgrounds). Check traps weekly. Traps can identify the pest before other signs are seen.
- Look for brown, silvery adult moth. This can be used as an early warning for *Tuta* infestation and the captured male adult moths show the pest population.
- Take action if 5-10 leaves across 500-100 plants are with larvae or tunnels.
- It is too late to control the pest if more than 30% of the fruits are infested with the larva.

Direct Control

- Crush larvae by hand once signs of tunnelling are observed. Remove all the leaves and fruits with tunnels on them to kill the larvae.
- Bury infested fruits and foliage at 50-100cm, to destroy any larvae or pupae present.
- Apply Neem products to the soil, the upper surface of leaves or directly on larvae to kill them. For homemade solutions, Neem seeds/leaves are mostly found in the valleys such as Chirundu, Siavonga and Luangwa.

Direct Control / Restrictions

Direct Control	Restrictions
• When using a pesticide or botanical, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, pre-harvest interval (PHI), max. number of sprays, and restricted re-entry interval (REI). Do not empty into drains and water sources.	
• Always consult a recent list of registered pesticides (e.g. ZEMA).	
• Only apply products when verified by monitoring.	
• Spray in the evening directly on the plants to kill the larvae.	
• *Bacillus thuringiensis var kurstaki* products; foliar sprays against young larvae.	• WHO class III (slightly hazardous); stomach poison for insects, must be eaten by insect; PHI 1 day, REI 1 day after spray, retreatment interval 7 days, max. 2 sprays.
• Nimbicidine: Azadirachtin-based products; foliar spray against young larvae	• Not WHO classified (unlikely to present acute hazard in normal use), moulting disruptor pesticide. PHI 3 d; REI 1 d. Max. 1 spray during flowering. Toxic to bees.
• Spinosad-based products; foliar spray against larvae.	• WHO class III (slightly hazardous); biological spinosyn pesticide. PHI 3 days, REI 1 day, min. retreatment interval 7 days, max. 2 sprays. Spray on days without rain, but not in bright sunlight.
• Flubendiamide (BELT 480SC at rate of 4mls per 20L).	• WHO class III (slightly hazardous). PHI 3 days, REI 1 day.

Ministry of Agriculture

Zambia

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